

Notes: Gross and Souleles (QJE, 2002)

Do Liquidity Constraints and Interest Rates Matter for Consumer Behavior? Evidence from Credit Card Data

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1 Big picture

1.1 Main question

Gross and Souleles ask whether household borrowing responds to changes in the supply and price of credit. The paper studies two central questions in consumption and household finance. First, when a credit card issuer increases an account's credit limit, does the account-holder actually borrow more? Second, when the issuer changes the account-specific interest rate, does debt fall when the price of credit rises?

1.2 Core answer

The answer is yes on both margins. Credit-limit increases produce an immediate and statistically significant increase in credit card debt. The long-run marginal propensity to borrow out of extra liquidity is roughly 10–14%, and it is much larger for accounts that begin close to their credit limit. Interest rates also matter: a one percentage point increase in the credit card rate reduces debt by about \$113 on the card over the next nine months. The authors argue that these patterns reject the canonical permanent-income benchmark with no liquidity constraints and support a view in which liquidity constraints and precautionary motives are quantitatively important.

1.3 Why credit card data are useful

The paper's identification comes from a unusually rich panel of credit card accounts. Unlike standard household surveys, these account data separately observe the credit limit, the balance, the interest-incurring debt, the interest rate, internal and external credit scores, account age, issuer policies, and other credit bureau balances. This separation is crucial because the authors can distinguish credit supply from credit demand: the credit limit and interest rate are supply-side objects, while debt is the household's borrowing response.

1.4 Economic interpretation

A credit-limit increase is not a wealth shock. It raises access to liquidity without directly increasing permanent income or net worth. Under a strict permanent-income model with no binding or anticipated liquidity constraint, debt should not respond to a predictable increase in available credit. A positive response therefore reveals either current constraints, precautionary responses to future possible constraints, or behavioral mechanisms that make credit availability itself affect spending and borrowing.

2 Incremental contribution of the paper

2.1 Direct evidence on credit supply rather than indirect proxies

Earlier tests of liquidity constraints often infer constraints from income, wealth, or asset-holding splits. Gross and Souleles study direct changes in credit supply: changes in credit limits and account-specific interest rates. This is a cleaner test because the treatment variable is the actual borrowing capacity or borrowing price faced by the account-holder.

2.2 Separation of credit limits, balances, and rates

The paper uses issuer account records that separately record limits, debt, and rates. This allows a direct estimate of the “MPC out of liquidity,” dD/dL , and the interest-rate sensitivity of debt, dD/dr . It also allows the authors to compare the response on the focal card with balances on other cards, which helps separate net borrowing from balance shifting.

2.3 Dynamic event-study design with strong controls

The main empirical design is a distributed-lag event study. It traces the cumulative response of debt month by month after a credit-limit or interest-rate change. The paper adds month fixed effects, account-level differencing, fixed account effects in some specifications, credit scores, account age, debt controls, manual-versus-automatic line-change controls, and instrumental variables based on issuer timing rules.

2.4 Heterogeneity as a test of mechanisms

A major contribution is not just the average response but the cross-sectional pattern. The debt response to credit-limit increases is largest for accounts already near the limit, which is exactly where quantity constraints should bind. At the same time, even low-utilization accounts respond, which points beyond the simplest binding-constraint story and toward buffer-stock or precautionary-saving logic.

2.5 A behavioral puzzle beyond conventional models

The paper shows that many credit card borrowers also hold low-yield liquid assets. This is hard to reconcile with standard no-arbitrage reasoning because borrowers could reduce expensive credit card debt by drawing down checking and savings balances. The finding motivates behavioral explanations such as self-control, mental accounting, or other departures from conventional optimization.

3 Data and empirical design

3.1 Account-level data

The data come from several anonymous U.S. bankcard issuers and cover roughly 24,000 accounts from January 1995 through January 1998. The main regressions use more than 200,000 account-month observations. The unit of observation is a credit card account, not a household. For many analyses, the authors use credit bureau information to observe balances on other credit cards held by the same borrower.

3.2 Key variables

- **Debt** is interest-incurring debt, not merely transactions balances.
- **Credit limit** is the account’s borrowing line at the start of the period.
- **Interest rate** is the account-specific annual percentage rate, including promotional rates.

- **Utilization** is balances divided by the credit limit.
- **Manual line changes** are requested by the consumer; **automatic line changes** are initiated by the issuer.
- **Credit scores** summarize issuer information about risk and profitability and are central controls for endogeneity.

3.3 Credit-limit specification

The main specification for credit limits is

$$\Delta D_{i,t} = \alpha' time_t + \beta_0 \Delta L_{i,t} + \beta_1 \Delta L_{i,t-1} + \cdots + \beta_{12} \Delta L_{i,t-12} + \gamma' X_{i,t} + \varepsilon_{i,t}.$$

The cumulative response after k months is $b_k = \sum_{j=0}^k \beta_j$. The long-run effect $b_{Tot} = b_{12}$ is interpreted as the fraction of a credit-line increase that is borrowed, or the MPC out of liquidity.

3.4 Interest-rate specification

The interest-rate specification is analogous:

$$\Delta D_{i,t} = \alpha' time_t + \beta_0 \Delta r_{i,t} + \beta_1 \Delta r_{i,t-1} + \cdots + \beta_9 \Delta r_{i,t-9} + \gamma' X_{i,t} + \varepsilon_{i,t}.$$

Here $b_{Tot} = b_9$ is the long-run dollar change in debt from a one percentage point increase in the credit card interest rate.

3.5 Endogeneity controls

The main concern is that issuers might increase limits or cut rates when they expect demand for credit to rise. The paper responds in several ways. Month dummies absorb aggregate credit cycles and seasonality. First differences remove fixed account-level debt and line levels. Manual line-change controls address consumer-requested credit expansions. Internal and external credit scores capture issuer information. Finally, IV specifications exploit mechanical timing rules, such as the number of months since the last credit-line or rate change.

4 Main findings

4.1 Credit limits affect borrowing

The average long-run response of debt to a limit increase is about 0.126, meaning that a \$1,000 increase in available credit raises debt by about \$126. For automatic issuer-initiated line changes, the estimate is still 0.096. The IV estimates are similar in magnitude. This is strong evidence that extra liquidity affects borrowing, inconsistent with the prediction that predictable credit-limit changes should have no effect under an unconstrained permanent-income model.

4.2 Interest rates affect borrowing

A one percentage point increase in the credit card rate lowers debt by roughly \$113 over nine months. The response is immediate and persistent. Debt also responds asymmetrically: rate decreases have a much larger effect than rate increases, which helps explain why issuers use promotional rate cuts to stimulate borrowing. The positive response of other-card balances indicates some balance shifting, but the main results imply that most of the response is a net reduction in total borrowing rather than merely moving balances across cards.

4.3 High-utilization accounts reveal binding constraints

Accounts starting above 90% utilization have an MPC out of liquidity of about 0.45, much larger than the responses of lower-utilization accounts. This is the most direct evidence of binding liquidity constraints: when borrowers are close to the credit limit, additional liquidity is quickly used.

4.4 Low-utilization accounts still respond

Accounts starting below 50% utilization have a statistically significant response of about 0.068. This is too large for a model in which only currently binding constraints matter. The paper interprets this as consistent with buffer-stock models, in which consumers maintain a target buffer of unused credit against future shocks.

4.5 Credit scores and demographics matter

Low-score accounts are far more likely to be near the credit limit and have a larger MPC out of liquidity than high-score accounts. Young account-holders and newer accounts also tend to have higher utilization and larger debt responses. These patterns support the idea that liquidity constraints are more severe for borrowers with weaker access to outside credit.

4.6 Credit card use creates a portfolio puzzle

The SCF evidence shows that many bankcard borrowers hold meaningful liquid assets while paying high credit card interest rates. This is difficult to reconcile with standard models because paying down credit card debt often earns a risk-free return equal to the credit card rate. The pattern suggests that liquidity constraints explain much of borrowing behavior, but not all of it.

5 Extracted figures/tables and interpretations

SAMPLE STATISTICS		
Variable	Mean	Median
debt	1349	70
debt debt > 0	2809	2120
credit limit	6207	5000
Δ credit limit	76.8	0
Δ credit limit	1985	1000
Δ credit limit $\neq 0$		
interest rate	16.6	17.2
debt-weighted	17.0	17.7
Δ interest rate	.036	0
abs(Δ interest rate)	.914	.250
Δ interest rate $\neq 0$		
# observations	231644	

The unit of observation is a credit card account. Debt is interest-incurring balances, as defined in the text. The differenced variables represent monthly changes. Debt and credit limits are measured in current dollars (1995–1998). The sample is that used for Table II, except for the interest rate statistics which correspond to Table III (# obs = 185,151).

Table I: Sample statistics.

Interpretation. Table I introduces the account-level dataset. The mean debt is \$1,349, but the median is only \$70, showing that many account-months have little or no revolving debt while a subset carries substantial balances. Conditional on having positive debt, mean debt is \$2,809 and median debt is \$2,120. The average credit limit is \$6,207, and the average interest rate is 16.6%, so this is a high-cost marginal source of credit. The table also shows that credit-limit and interest-rate changes are lumpy: most monthly changes are zero, but when limits change, the average nonzero change is about \$1,985.

THE RESPONSE OF DEBT TO INCREASES IN THE CREDIT LIMIT				
Row		b_{Tot}	S.e.	# obs
(1)	average MPC (dD/dL)	0.126	0.021	231644
(2)	automatic dL	0.096	0.016	145429
	manual dL	1.673	0.773	
(3)	fixed account effects	0.138	0.029	145429
(4)	credit scores	0.085	0.021	127431
(5)	scores, debt, account age	0.122	0.022	131636
(6)	IV	0.111	0.018	145396
(7)	IV: scores, account age	0.083	0.030	131603
(8)	IV: $\#(dL \neq 0)$	0.104	0.020	145396
(9)	d (interest rates)	0.122	0.024	137279
(10)	interest rates	0.122	0.024	137279
(11)	balances on other cards	0.026	0.030	130486

A distributed lag model (equation (1)) was used to estimate the dynamic response of credit card ΔD to changes in credit limits ΔL including twelve lags. b_{Tot} ($=b_{12}$) gives the long-run, cumulative change in debt as a fraction of the change in the line (the MPC out of liquidity), dD/dL . All regressions include a full set of month dummies. The standard errors allow for heteroskedasticity across accounts as well as serial correlation within accounts. Sample sizes vary with missing variables. Row (2) distinguishes manual line changes (requested by the consumer) from automatic changes (initiated by the issuer), and includes separate intercepts for each case. Rows (3)–(8) include the controls for the manual line changes; the reported b_{Tot} is for the automatic changes. Row (3) includes a fixed effect by account. Row (4) includes as controls a cubic polynomial in the normalized credit scores, both internal and external, interacted by issuer dummies; and then twelve lags of all these terms (i.e., 2 scores $\times$ 3 polynomial terms $\times$ 13 lags $\times$ issuer dummies). Row (5) includes cubic polynomials in the two scores, debt and account age; all from month $t - 1$ and all interacted by issuer dummies. Rows (6)–(8) instrument for ΔL with indicator variables for the number of months since the latest change in line. Row (7) includes as controls cubic polynomials in the two scores and account age, from month $t - 1$. Row (8) includes dummy variables for the total number of line changes each account received during the sample period. Row (9) includes as controls the change in account interest rate with twelve lags. Row (10) includes instead the level of the interest rate with twelve lags. In row (11) the dependent variable is balances on *other* credit cards held by the account-holder.

Table II: Response of debt to increases in the credit limit.

Interpretation. Table II is the central test of whether liquidity matters. Row (1) estimates an average MPC out of liquidity of 0.126, so a \$1,000 increase in the credit line raises debt by about \$126. Row (2) shows that automatic issuer-initiated increases still raise debt by 0.096 per dollar of extra line, while manual consumer-requested increases have a much larger coefficient. Rows with fixed effects, credit scores, debt, account age, and IV timing instruments preserve the result. The final row shows only a small and insignificant response of balances on other cards, suggesting that extra limits mainly increase borrowing on the treated card rather than simply reshuffling balances.

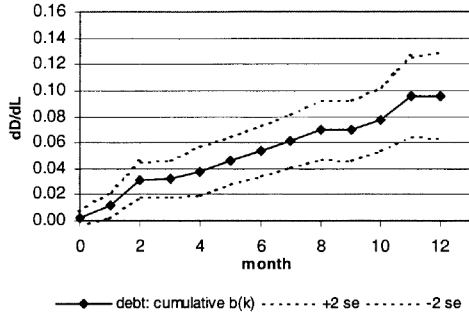


FIGURE I
The Cumulative Response of Debt to “Automatic” Increases in the Credit Line,
per Dollar of Extra Line (Table II, Row (2))

Figure I: Cumulative response of debt to automatic credit-line increases.

Interpretation. Figure I visualizes the dynamics behind Table II, row (2). Debt begins rising immediately after an automatic line increase and continues to rise over the following year, approaching an MPC near 0.10 by month 12. The smooth upward path is important because it looks like a real behavioral response rather than a one-month accounting artifact. The dotted confidence bands remain mostly above zero after the early months, reinforcing the claim that liquidity changes have economically meaningful effects on debt.

THE RESPONSE OF DEBT TO CHANGES IN INTEREST RATES

Row		b_{Tot}	S.e.	# obs
(1)	interest sensitivity (dD/dr)	-112.6	8.4	185151
(2)	normalize by line: $d(D/L)/dr$	-0.016	0.001	185151
(3)	fixed account effects	-132.4	7.9	185151
(4)	credit scores	-109.4	11.9	107542
(5)	scores, debt, account age	-117.1	9.5	173406
(6)	IV	-79.1	27.2	147158
(7)	IV: scores, account age	-156.4	32.1	138709
(8)	IV: $\#(dr \neq 0)$	-80.8	26.8	147158
(9)	$d(\text{credit limit})$	-111.8	-8.3	185099
(10)	increase r	-90.1	9.2	185151
	decrease r	-317.7	63.7	
(11)	balances on other cards	37.8	20.2	142334

A distributed lag model (equation (2)) was used to estimate the dynamic response of credit card debt ΔD to changes in credit card interest rates Δr including nine lags. b_{Tot} ($=b_9$) gives the long-run, cumulative change in debt per percentage point increase in rates, dD/dr . All regressions include a full set of month dummies. The standard errors allow for heteroskedasticity across accounts as well as serial correlation within accounts. Sample sizes vary with missing variables. In row (2) only, the dependent variable is the change in the ratio of debt to the credit limit. Row (3) includes a fixed effect by account. Row (4) includes as controls a cubic polynomial in the normalized credit scores, both internal and external, interacted by issuer dummies; and then nine lags of all these terms (i.e., 2 scores $\times$ 3 polynomial terms $\times$ 10 lags $\times$ issuer dummies). Row (5) includes cubic polynomials in the two scores, debt and account age; all from month $t-1$ and all interacted by issuer dummies. Rows (6)–(8) instrument for Δr with indicator variables for the number of months since the latest change in rates. Row (7) includes as controls cubic polynomials in the two scores and account age, from month $t-1$. Row (8) includes dummy variables for the total number of rate changes each account received during the sample period. Row (9) includes as controls the change in account credit limit with nine lags. Row (10) distinguishes the response to increases versus decreases in interest rates, including separate intercepts for each case. In row (11) the dependent variable is balances on *other* credit cards held by the account-holder.

Table III: Response of debt to changes in interest rates.

Interpretation. Table III shows that debt is highly sensitive to the account-specific price of credit. A one percentage point rate increase lowers debt by \$112.6 in the baseline specification, and the result remains robust to account fixed effects, credit scores, account characteristics, and IV timing instruments. Row (10) shows strong asymmetry: debt responds much more to rate decreases than to rate increases, consistent with promotional low-rate offers expanding borrowing. Row (11) shows other-card balances rise by \$37.8 when the focal card rate rises, so some borrowers shift balances away from the expensive card, but balance shifting explains less than half of the own-card response.

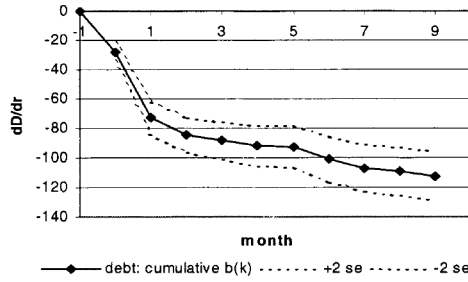


FIGURE II
The Cumulative Response of Debt to Increases in the Interest Rate, per Percentage Point (Table III, Row (1))

Figure II: Cumulative response of debt to interest-rate increases.

Interpretation. Figure II shows that the interest-rate response is both fast and persistent. Debt falls sharply in the first two months after a rate increase and continues to drift downward through month 9, ending near the Table III estimate of about -\$113 per percentage point. The response is not a delayed or fragile pattern; it appears immediately and remains statistically clear. This figure is central because many earlier studies found small interest-rate effects, while the account-level data reveal a strong effect once the actual marginal card rate is observed.

UTILIZATION RATES AND LIQUIDITY CONSTRAINTS

Row	b_{Tot}	S.e.	# obs
A. Credit limit changes			
(1) dD/dL			143511
utilization < .50	0.068	0.018	
utilization .50-.90	0.158	0.060	
utilization > .90	0.452	0.125	
(2) $d(utilization)/dL$			143511
utilization < .50	-0.012	0.004	
utilization .50-.90	0.013	0.015	
utilization > .90	0.012	0.026	
B. Interest rate changes			
(3) dD/dr			182317
increase r			
utilization < .50	-65.0	11.4	
utilization .50-.90	-129.5	20.5	
utilization > .90	-85.5	21.4	
decrease r			
utilization < .50	-296.9	99.7	
utilization .50-.90	-429.3	120.2	
utilization > .90	-151.9	63.7	

This table contrasts the response of debt to credit supply (b_{Tot}) across accounts starting with different initial utilization rates (balances divided by the credit limit). In Panel A, for changes in credit limits (equation (1)), utilization is taken from month $t - 13$; in Panel B, for changes in interest rates (equation (2)), it is taken from month $t - 10$. Row (3) also distinguishes increases versus decreases in interest rates, for each utilization group. In rows (1) and (3) the dependent variable is the change in debt ΔD . In row (2) it is the change in utilization rate. Panel A includes controls for manual credit limit changes; the reported b_{Tot} is for the automatic changes. All regressions include separate intercepts for each utilization group, and a full set of month dummies. The standard errors allow for heteroskedasticity across accounts as well as serial correlation within accounts.

Table IV: Utilization rates and liquidity constraints.

Interpretation. Table IV is the key heterogeneity test. In Panel A, accounts starting below 50% utilization borrow 6.8 cents per extra dollar of credit, accounts between 50% and 90% borrow 15.8 cents, and accounts above 90% borrow 45.2 cents. This monotonic pattern strongly supports binding liquidity constraints near the limit. Row (2) shows that utilization itself changes little after line increases, which is consistent with buffer-stock behavior: borrowers may treat unused credit as a precautionary buffer. Panel B shows that high-utilization accounts are less interest-rate sensitive than middle-utilization accounts, also consistent with constraints limiting the ability to adjust debt when rates change.

Row	(1) % with utilization > .90	(2) Credit limit MPCs $\Delta D/\Delta L$		(3) Interest rate sensitivities $\Delta(D/L)/\Delta r$	
		b_{Fut}	S.e.	b_{Fut}	S.e.
(1) Credit limit					
small ($\leq \$5000$)	16.9	0.081	0.035	-0.016	0.002
large ($> \$5000$)	9.2	0.100	0.023	-0.017	0.002
p-value		0.675		0.512	
N		145429		185151	
(2) Credit score					
low (≤ 5)	25.8	0.125	0.026	-0.016	0.002
high (> 5)	1.8	0.070	0.020	-0.020	0.002
p-value		0.066		0.059	
N		132121		169234	
(3) Income					
low ($\leq \$31000$)	14.0	0.122	0.049	-0.015	0.002
high ($> \$31000$)	12.6	0.114	0.019	-0.018	0.009
p-value		0.876		0.103	
N		116542		141675	
(4) Age of account-holder					
young (≤ 43)	17.7	0.123	0.033	-0.020	0.002
old (> 43)	11.5	0.068	0.027	-0.017	0.002
p-value		0.182		0.196	
N		91832		132243	
(5) Age of account (years since booking)					
new (≤ 4)	14.7	0.103	0.018	-0.017	0.001
old (> 4)	12.9	0.067	0.029	-0.015	0.002
p-value		0.259		0.365	
N		145429		185151	
sample average	13.8				

This table contrasts the response of debt to credit supply (b_{Fut}) across accounts of different characteristics. In column (2) the dependent variable is the change in debt ΔD , and the independent variable are changes in the credit limit ΔL (equation (1)). In column (3) the dependent variable is the change in debt normalized by the line and the independent variables are changes in interest rates Δr (equation (2)). Each row in column (2) and in column (3) represents a separate regression. Column (1) reports the fraction of accounts with utilization rates above 90 percent. Each row applies the sample at approximately the median of the characteristics in that row. For the time-varying characteristics in rows (1) and (2), the characteristics are taken from month $t-15$ for credit limits (column (2)), month $t-10$ for interest rates (column (3)). For the characteristics in rows (3)-(5), which come from cross-sectional application data, their 1995 values are used. Row (2) uses the normalized interest credit score. Income is real application income (1982-1988), extrapolated to 1995 assuming a real growth rate of 2.5 percent per year. The reported p-values are for tests that b_{Fut} is the same across both groups in the row. Except for row (3), column (2) includes controls for annual credit limit changes; the reported b_{Fut} is for the automatic changes. See text for further details. All regressions include separate intercepts for each characteristic group, and a full set of month dummies. Sample sizes vary with missing variables. The standard errors allow for heteroskedasticity across accounts.

Table V: Heterogeneity in credit-card usage.

Interpretation. Table V connects borrowing responses to borrower characteristics. Low credit-score accounts are much more likely to be near the limit: 25.8% have utilization above 90%, compared with only 1.8% of high-score accounts. Low-score borrowers also have a higher MPC out of liquidity, 0.125 versus 0.070, with a p-value of 0.066. Young borrowers also have higher high-utilization rates and larger MPCs than older borrowers. These patterns show that liquidity constraints are not uniform across consumers; they are strongest among borrowers with weaker credit access or earlier life-cycle positions.

6 Interpretation of the main evidence

6.1 What rejects the unconstrained permanent-income benchmark

The main rejection is not simply that some people hold credit card debt. The sharp evidence is that debt rises after predictable increases in credit limits and falls after account-specific interest-rate increases. Under an unconstrained permanent-income model, a credit-limit increase that does not change wealth should not cause an immediate rise in borrowing. The data show the opposite.

6.2 What supports binding liquidity constraints

The utilization split provides direct evidence. Borrowers near their limit react much more strongly to extra liquidity than borrowers far from their limit. This is the pattern one would expect if the credit limit is a real quantity constraint for high-utilization accounts.

Variable	(1) All borrowers			(2) By income		
	Mean	Median	% positive	% positive	\$15K-50K	>\$50K
Net worth	\$127980	\$61530	95	89	94	99
Net housing equity	40568	22700	69	54	63	86
Total assets	180657	112850	100	98	100	100
Financial assets	47346	11700	98	93	98	100
Nonretirement financial assets	26546	5450	98	93	98	100
Liquid assets	5536	1800	96	91	96	98
Liquid assets - one month's income	1671	-385	33	46	30	34
% with bankcards	67			32	71	94
% borrowinghave bankcards	56			56	59	52

Variable	(3) By age			(4) By education		
	<35	% positive 35-62	>62	<HS	% positive HS	COL
Net worth	89	97	100	95	95	95
Net housing equity	43	79	80	77	69	67
Total assets	100	100	100	99	100	100
Financial assets	98	98	97	95	98	99
Nonretirement financial assets	98	98	97	95	98	99
Liquid assets	95	96	97	91	96	99
Liquid assets - one month's income	25	32	54	35	31	36
% with bankcards	59	72	64	35	65	88
% borrowinghave bankcards	69	64	29	54	63	49

Source: 1995 SCF, N = 3258. Results are weighted to be representative of the United States. The bottom two rows give the fraction of households in each group (each column) that have bankcards, and the fractions that are borrowing on their bankcards conditional on having bankcards. Borrowing households are those rolling over positive bankcard balances after their latest payments. The column labeled "% positive" gives the fraction of households in the group whose holdings of the asset in that row are positive. Asset definitions generally follow standard SCF textbook suggestions, with the exception that net worth does not include credit card debt. Nonretirement financial assets exclude IRAs and defined-contribution pensions. Liquid assets include accounts in checking, saving, money-market, and money-market accounts. Income is gross total household income. Asset holdings are all conditional on having and borrowing on at least one bankcard. Age and education refer to the household head. Columns (2) and (3) contrast the bottom 25 percent, the middle 50 percent, and the top 25 percent of the income and age distributions, respectively.

Table VI: Asset holdings of bankcard borrowers.

Interpretation. Table VI documents the portfolio puzzle. Among households borrowing on bankcards, 95% have positive net worth, 96% have positive liquid assets, and one-third have liquid assets exceeding one month's income. This is puzzling because credit card interest rates are high, and liquid assets usually earn low returns. The bottom rows also show that 67% of households have bankcards and, conditional on having one, 56% borrow. Borrowing is especially common among young households, but even high-income and college-educated households often borrow on bankcards. The table therefore pushes the paper beyond standard liquidity-constraint models toward behavioral or portfolio-friction explanations.

6.3 Why precautionary motives are needed

Even low-utilization borrowers respond to credit-line increases. A model in which only currently constrained borrowers respond cannot explain this. Buffer-stock models can: consumers may target a desired buffer of unused credit, so an increase in the line permits more borrowing while still preserving a precautionary cushion.

6.4 Why interest-rate results matter

The interest-rate results show that consumers are not completely insensitive to credit prices. This matters for monetary transmission, credit card pricing, promotional offers, and the welfare effects of consumer credit regulation. The fact that rate decreases produce especially large borrowing responses helps explain why issuers use temporary teaser rates and promotional balance-transfer offers.

6.5 Why the asset-holding puzzle matters

The coexistence of credit card borrowing and liquid asset holdings is difficult to reconcile with simple optimization. It suggests that a complete model of consumer credit behavior may need to combine liquidity constraints and precautionary motives with behavioral frictions such as mental accounting, self-control, or bankruptcy considerations.

7 Critical evaluation

7.1 Strengths

- The data are unusually clean: the paper observes actual credit limits, interest rates, balances, credit scores, and account histories.
- The distributed-lag design traces the dynamic response rather than only comparing before and after averages.
- The authors address endogeneity with multiple strategies, including issuer timing instruments.
- The heterogeneity by utilization provides a direct mechanism test for liquidity constraints.
- The paper distinguishes own-card responses from balance shifting across other cards.

7.2 Limitations

- The unit of observation is an account, not a full household balance sheet. The authors use bureau data and SCF data to mitigate this, but some household assets and liabilities remain unobserved.
- The paper studies debt rather than consumption directly. More borrowing is interpreted as spending or debt-financed consumption, but some borrowing could finance asset reallocation or other uses.
- The identifying variation comes from credit card accounts in the mid-1990s, so magnitudes may differ in other credit regimes.

- The issuer timing instruments are compelling, but their validity depends on timing rules being unrelated to high-frequency credit demand shocks after controls.

7.3 Bottom line

Gross and Souleles provide strong micro evidence that liquidity constraints and interest rates matter for consumer debt. The paper's most important message is quantitative: credit supply shocks have large effects, especially for borrowers near their limits, and these effects are large enough to matter for aggregate credit cycles and monetary transmission.